# How the tool estimates a repeat's combined length (both alleles) from short reads

**Input: short-read BAM/CRAM**

30x Illumina, aligned to GRCh38
(training data: 1000 Genomes NYGC CRAMs)

**Repeat annotation (per locus)**

motif, reference length, GC, segmental dup.
known repeats only: long-read panel
statistics (mean, core spread, outliers)

**Training truth (not needed to run)**

vamos long-read genotypes: combined
repeat units per person, 290 people
(training, calibration and testing)

## 1 Read-level signals, extracted for every repeat

- read depth over the repeat vs flanking sequence, at 4 mapping-quality levels
- reads made of the repeat's motif (in-repeat reads, anchored pairs)
- read pairs spanning the repeat: fragment-length shift and spread
- divided by the person's genome-wide depth, fragment size and read length

*is the repeat in the catalog?*

## 2a Known repeats (in the long-read catalog)

Per-repeat calibration curve: monotone (isotonic) fit
of truth on the best signal, robust-linear (Theil-Sen)
beyond the training range; fitted per repeat

Anchor = this curve's estimate
Features = signals + annotation + panel statistics
+ curve and transfer-model estimates

## 2b New repeats (not in catalog)

No per-repeat curve or panel statistics

Anchor = 2 x reference copy number
Features = signals + reference-derived
annotation + transfer-model estimate
(gradient-boosted trees trained across
all repeats)

## 3 Gradient-boosted trees model the correction to the anchor

target = log(true + 1) - log(anchor + 1): one model for the point estimate (squared error),
two for the 5th and 95th percentiles (quantile loss)

4 Split-conformal calibration on held-out data (validation people for known repeats, validation repeats
for new ones): the 5-95% range is widened or narrowed per repeat group so 90% of true values fall inside,
then rounded out to whole repeats

## Output per repeat

**Point estimate**

combined repeat units
(both alleles), decimal

**90% interval**

whole repeat units;
contains the truth for
91-92% of test calls

**Tier**

PASS: interval narrower
than the estimate and
repeat not unusual
LOW_CONF: otherwise
NA: no usable reads